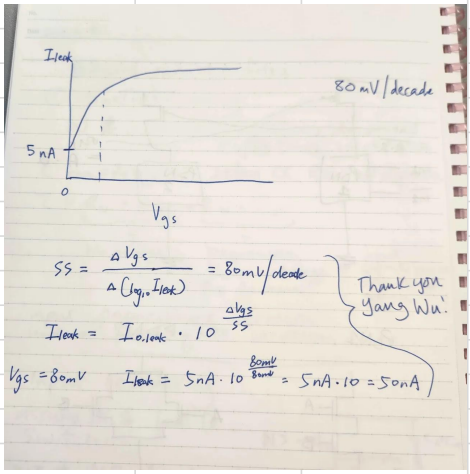
3A

Subthreshold slope		80mv/decade
Vgs	Leakage current	
0	5nA	



I apologise for the trouble! Due to my disability, my wrists and hands are not working well at the moment so I must type all my work. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.



SS = Subthreshold Slope

Voltage Vgs	Original I Leak	SS	**Leakage current**	Nominal Case
0	5	80	**5**	
80	5	80	**50**	
160	5	80	**500**	
240	5	80	**5000**	
mV	nA	mV/decade	nA	

3B

Vds = Vgs-Vth.

If Vth is lowered by 80mV, and the effective value of Vds remains the same, then Vgs is effectively increased by 80mV. This is because the leakage current formula depends on the difference between Vth and Vgs, not strictly Vgs (?).

Voltage Vgs	Added 80mV	Original I Leak	SS	**Leakage current**	Threshold Shift
0	80	5	80	**50**	
80	80	5	80	**500**	
160	80	5	80	**5000**	
240	80	5	80	**50000**	
mV	mV	nA	mV/decade	nA	

3C

Voltage Vgs	Original I Leak	SS	Leakage current	Nominal Case
0	5	80	5	
mV	nA	mV/decade	nA	

50000	devices
250000	total block current
0.9	supply voltage
225000	Power
2.25mW	

Voltage Vgs	Added 80mV	Original I Leak	SS	Leakage current	Threshold Shift
0	80	5	80	50	
mV	mV	nA	mV/decade	nA	

	50000	devices
	2500000	total block current
	0.9	supply voltage
	2250000	Power
	22.5mW	
3D	A designer might intentionally use lower threshold devices on a critical path despite the leakage penalty to reduce the propagation delay from changing inputs. This is important because a critical path, the longest path possible, is necessarily the bottleneck for maximum speed of the circuit, and so reducing this path by any means necessary benefits the block as a whole.	
3E	This is absolutely not correct - the threshold shift changes the leakage by a power of 10 with each shift of 80mV! at tiny voltages, that might be small, but the change in leakage grows so quickly.	